

X402SCOPE

X402SCOPE - Qinying Wang, Yong Yang, Yuan Chen, Shouling Ji, Mathias Payer, HexHive, EPFL.

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- Repository: <<https://github.com/HexHive/x402scope>>
- Commit: 8992d71e1795c8ef590e07682379b0a2b6acd255
- Documents preserved: 2

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README.md

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When HTTP 402 Meets the Blockchain: Risks on Emerging x402 Payments

This repo contains the **core code for x402scope**, including the **on-chain measurement code referenced in the paper** and **security rule-violation detection scripts**. It provides measurement scripts for collecting and analyzing x402-related on-chain data, plus security rule-violation checks. It does not include attack or exploit tooling.

For artifact evaluation, evaluators should follow the fixed workflows and commands specified in this Appendix and download the restricted-access version from Zenodo. The Appendix defines the version-specific AE workflow, while the repository README provides broader instructions for reuse, extension, and testing additional or updated facilitator deployments.

Repository Structure

- `Measurement/` : Core scripts for blockchain data ingestion (ETL), database management, and statistical analysis.
- `Measurement/paper_figdata/` : Scripts specifically tuned to generate the data points for Figures and Tables in the paper.
- `SecurityViolation/` : core code for x402scope for detecting and verifying security rule violations against live x402 facilitators.
- `requirements.txt` : Python dependencies.
- `config.example.py` : Example template for configuration and API keys.

1. Prerequisites

Hardware & OS

- **OS**: Linux (Ubuntu 22.04+ recommended).
- **Python**: Version 3.10 or higher.
- **Storage**: At least **1TB SSD** is recommended if reproducing the full blockchain data ingestion (Base block history and Solana transaction history).
- **Database**: A running **MariaDB** (or MySQL) instance.

External Services

To reproduce the measurement results, you require access to high-performance RPC nodes. The scripts perform heavy historical data fetching.

- **Base RPC:** An **Alchemy Base Mainnet** endpoint is required for stable block and receipts downloading.
- **Solana RPC:** We recommend an **Alchemy Solana Archival Node**.

2. Installation & Setup

Environment Setup

Clone the repository and install dependencies:

```
git clone <repository-url>
cd x402scope

# Create virtual environment
python3 -m venv .venv
source .venv/bin/activate

# Install dependencies
pip install -r requirements.txt
```

Configuration

1. Copy the template to a local configuration file:

```
cp config.example.py config.py
```

1. Edit `config.py` :
 2. **Database:** Fill in `MYSQL_USER` , `MYSQL_PASSWORD` , `MYSQL_HOST` , and `MYSQL_DB` .
 3. **RPC Endpoints:** Insert your Alchemy RPC urls for Base and Solana.
 4. **Private Keys:** Provide your Base and Solana private keys for executing rule checks in `config.py` .
 5. **Facilitators:** the included template keeps Coinbase fields (`CDP_API_KEY_ID` , `CDP_API_KEY_SECRET`) and `ThirdWeb_Secret_key` ; leave unused credential fields blank.
1. Copy/Link the `config.py` to `SecurityViolation` folder:

```
cp config.py SecurityViolation/config.py
```

Recommended targets and test-account setup

We recommend starting with the bounded Coinbase test targets instead of running the full facilitator list:

Target	Network	Recommended first test
coinbase-test	Base Sepolia / EVM	Start <code>server_app_v2.py</code> , then run <code>evmclient_v2.py</code>
coinbase-solanadev	Solana Devnet	Run <code>verify_settle_solanav2.py</code>

Test accounts are required. Do **not** commit private keys.

1. Generate EVM test accounts for Base Sepolia.

Use an EVM wallet such as Rabby, MetaMask, or another compatible wallet to create/export the test-account private keys. For the full EVM verify/settle test, prepare:

- `pk1` : funded client/payer private key. For Base Sepolia-only tests, reviewers may use the shared deterministic test key `"1" * 64` , whose address is `0x19E7E376E7C213B7E7e7e46cc70A5dD086DAff2A` , after funding it with testnet USDC/gas.
- `pk2` : merchant/server recipient private key
- `poor_pk` : unfunded or underfunded client private key used for negative checks

Export any private keys locally and place them only in your private `config.py` . The `"1" * 64` key is only for public testnet experiments and must not be used on mainnet.

1. Generate Solana Devnet test accounts.

The Solana scripts expect base58-encoded private keys in `config.py` . Use the helper script to generate a keypair:

```
python3 SecurityViolation/Scripts/generate_solana_keypair.py -o solana-server.json
```

The command prints the Solana public address and the `Private key for config.py` value. It also writes a Solana CLI-style JSON byte-array key file to the `-o` path.

Generate the required Solana local keys and place the printed base58 private-key strings in `config.py` :

- `solpk` : funded client/payer private key
- `solpk_server` : merchant/server recipient private key
- `poor_pk_sol` : unfunded or underfunded client private key used for negative checks

1. Fund the test accounts and initialize Solana token accounts.

Use Circle's faucet to obtain testnet USDC:

- `<https://faucet.circle.com/>`
- choose **Ethereum Sepolia** for EVM test USDC used by the Base Sepolia flow
- choose **Solana Devnet** for Solana Devnet test USDC

If gas is needed, use the relevant network faucets:

- Base Sepolia gas: `<https://www.alchemy.com/faucets/base-sepolia>`
- Solana Devnet SOL: `<https://faucet.solana.com/>`

Alternatively, request Devnet SOL directly with:

```
python3 SecurityViolation/Scripts/fund_solana_devnet_sol.py \  
--to <target-solana-public-address> \  
--amount 1
```

This helper requires no private key, waits for confirmation, and reports the balance before and after the airdrop. The configured maximum per request is 5 SOL. If a public RPC returns HTTP 429 because its faucet quota is exhausted, wait and retry or pass another Devnet RPC with `--rpc`.

If a Solana USDC ATA is missing, create it with:

```
python3 SecurityViolation/Scripts/create_solana_usdc_ata.py \  
--network solana-devnet \  
--payer-key solpk \  
--owner-key solpk
```

The payer signs and pays the ATA rent. The helper is idempotent and does not send a transaction when the ATA already exists. Use `--owner-address` for an owner not listed in `SecurityViolation/config.py`, or `--yes` to skip confirmation.

1. Create Coinbase CDP API credentials.

Register at <https://portal.cdp.coinbase.com/> and create secret API keys. Put the generated values in your local `config.py`:

```
CDP_API_KEY_ID = "..."  
CDP_API_KEY_SECRET = "..."
```

1. Fill `config.py` and local target values.

Fill the local keys, RPC endpoints, and Coinbase credentials in `config.py`, then copy it to `SecurityViolation/config.py` as shown above. Also set `pay_to_address` for the selected target in `SecurityViolation/target.py` to the reviewer-controlled merchant/recipient address:

- `coinbase-test`: EVM recipient address derived from `pk2`
- `coinbase-solanadev`: Solana recipient address derived from `solpk_server`

1. Run the read-only preflight check.

Before running a state-changing security test, check the local environment and selected target.

```
python3 SecurityViolation/preflight_check.py -t coinbase-test  
python3 SecurityViolation/preflight_check.py -t coinbase-solanadev
```

To additionally check the selected target's read-only on-chain balances, add `--check-balances`. For Solana this includes native SOL balances, the existence and balance of each owner's USDC associated token account (ATA), and the configured fee payer's SOL balance:

```
python3 SecurityViolation/preflight_check.py -t coinbase-test --check-balances
python3 SecurityViolation/preflight_check.py -t coinbase-solanadev --check-balances
```

The EVM check reads `pk1` , `pk2` , and `poor_pk` . The Solana check reads `solpk` , `solpk_server` , and `poor_pk_sol` (also accepting the local alias `poor_pk_pool`). A missing ATA for the intentionally unfunded poor account is reported as a warning. These checks use the selected network's public RPC and never send a transaction.

The command reports missing dependencies, credentials, addresses, fee-payer settings, authorization timing, and the expected CAIP-2 network. It returns a non-zero exit status when a required item is missing. For reliable verify/settle checks, we recommend that `valid_before_offset` should be greater than 180 seconds.

For the free-shopping attack, the reviewer should temporarily reduce `valid_before_offset` to a small value (for example, `8`) and vary it to find the boundary where verify succeeds but settle fails. This short value is intentional for the attack and should not be treated as the normal default.

1. Run tests one by one.

For the current Coinbase CDP v2 endpoint, start with the v2 local merchant/client flow. Run the server in one terminal and the client in another terminal:

```
python3 SecurityViolation/server_app_v2.py -t coinbase-test
python3 SecurityViolation/evmclient_v2.py -t coinbase-test
```

Then run other direct checks as needed:

```
python3 SecurityViolation/verify_settle_base_v2.py -t coinbase-test
python3 SecurityViolation/verify_settle_solana_v2.py -t coinbase-solanadev
```

3. Experiment Workflow

Part A: Security Rule Verification (SecurityViolation)

This section reproduces the detection of security rules described in the paper.

1. **Define Config:** ensure your `config.py` includes:

- **Private keys:** `pk1` and `poor_pk` (test payer), `pk2` (recipient), `solpk` and `poor_pk_sol` (Solana payer), and `solpk_server` (recipient)
- **API credentials:**
 - Coinbase: `CDP_API_KEY_ID` , `CDP_API_KEY_SECRET` (for coinbase targets)
 - Thirdweb: `ThirdWeb_Secret_key` (only needed for thirdweb targets; keep it blank otherwise)

1. **Define Targets:** Ensure `target.py` contains the facilitator metadata (API endpoints, network, amount, timing window, and local merchant settings).

The target file provides facilitator metadata and local merchant settings. Merchant `pay_to_address` values are operator-provided test addresses: they are quick to generate and configure, are not intrinsic to x402scope, and are not required for inspecting the preserved evidence. To re-run checks, reviewers should insert their own merchant/test addresses. You may add or customize targets in `target.py` (FacilitatorTarget entries in `FACILITATORS`):

```
"custom-target": FacilitatorTarget(
    name="my_facilitator",
    facilitator_base="https://your-facilitator.example.com/x402",
    merchant_base="http://127.0.0.1:8001",
    network="base", # or "base-sepolia", "solana", "solana-devnet"
    price="$0.001",
    pay_to_address="0x...", # Your merchant/test address
    pay_amount=1000, # Amount in smallest unit
    threads=1,
    valid_after_offset=-60,
    valid_before_offset=300, # Recommended >= 180 for normal verify/settle checks
    description="my facilitator.",
)
```

1. **Execute Detection:** Run one x402scope test script against one specific target.

Because target configuration is facilitator-specific and may require different credentials, networks, fee-payer settings, and test-account balances, reviewers should **select one target and one test script at a time**.

Example direct invocations:

```

# Recommended first EVM/Base Sepolia Coinbase v2 flow:
# terminal 1: local merchant server
python3 SecurityViolation/server_app_v2.py -t coinbase-test

# terminal 2: EVM client flow
python3 SecurityViolation/evmclient_v2.py -t coinbase-test

# Additional EVM/Base Sepolia Coinbase v2 checks
python3 SecurityViolation/verify_settle_base_v2.py -t coinbase-test
python3 SecurityViolation/erc1271testv2.py -t coinbase-test
python3 SecurityViolation/erc6492testv2.py -t coinbase-test

# Legacy/v1-compatible EVM checks for v1-compatible targets
python3 SecurityViolation/verify_settle_base.py -t <v1_compatible_target>
python3 SecurityViolation/erc1271test.py -t <v1_compatible_target>
python3 SecurityViolation/erc6492test.py -t <v1_compatible_target>

# Solana Devnet Coinbase v2 target
python3 SecurityViolation/verify_settle_solanav2.py -t coinbase-solanadev

```

Current security test scripts:

Script	Chain / Scope	Purpose
verify_settle_base.py	EVM, legacy/v1	Verify/settle checks for legacy or v1-compatible EVM targets.
verify_settle_base_v2.py	EVM, x402 v2	Verify/settle checks for x402 v2 EVM targets.
verify_settle_solana.py	Solana, legacy/v1	Verify/settle checks for legacy or v1-compatible Solana targets; requires a configured fee payer.
verify_settle_solanav2.py	Solana, x402 v2	Verify/settle checks for x402 v2 Solana targets; requires a configured fee payer.
erc6492test.py	EVM, legacy/v1	ERC-6492 validation behavior test for legacy or v1-compatible targets.
erc6492testv2.py	EVM, x402 v2	ERC-6492 validation behavior test for x402 v2 targets.
erc1271test.py	EVM, legacy/v1	ERC-1271 signature-validation behavior test for legacy or v1-compatible targets.
erc1271testv2.py	EVM, x402 v2	ERC-1271 signature-validation behavior test for x402 v2 targets.
evmclient.py with server_app.py	EVM client flow, legacy/v1	Legacy client-flow test; start server_app.py before running evmclient.py .
evmclient_v2.py with server_app_v2.py	EVM client flow, x402 v2	Recommended Coinbase EVM client-flow test; start server_app_v2.py before running evmclient_v2.py .

Supporting scripts:

Script	Role
basesupport.py	Optional EVM pre-check for basic facilitator support.
server_app_v2.py	Local merchant server used by the x402 v2 EVM client-flow test.
server_app.py	Local merchant server used by the legacy/v1 EVM client-flow test.

Direct-script template:

```
python3 SecurityViolation/<script>.py -t <target_facilitator>
```

For the x402 v2 EVM client-flow test, run the local merchant server in one terminal, then run the client script in another terminal:

```
python3 SecurityViolation/server_app_v2.py -t <target_facilitator>
python3 SecurityViolation/evmclient_v2.py -t <target_facilitator>
```

For legacy/v1-compatible targets, use:

```
python3 SecurityViolation/server_app.py -t <target_facilitator>
python3 SecurityViolation/evmclient.py -t <target_facilitator>
```

Part B: Empirical Measurement (Base)

This workflow reproduces the data collection for the Base network (Block height > 30M, approx. 2025/05/09).

1. Ingest Raw Data:

Download blocks and filter for facilitator interactions.

```
cd Measurement/
python3 base_downloadblocks.py # Connects to Alchemy Base RPC
python3 x402fetchtxs.py # Filters txs by facilitator EOAs
```

1. Verify Integrity:

Ensure the nonce consecutiveness so that no blocks were skipped due to RPC timeouts.

```
python3 base_confirmintegrity.py
```

1. Process Metadata:

Fetch transaction receipts to determine success status, revert reasons, and gas usage.

```
python3 base_updatestatus.py
python3 base_updaterevertreason.py
```

1. Generate Statistics:

```
mysql -h <host> -p < base_stattable.sql
```

Part C: Empirical Measurement (Solana)

This workflow reproduces the data collection for the Solana network.

1. Fetch Transaction History:

Uses `getSignaturesForAddress` to sync facilitator history. *Note: This step benefits significantly from Alchemy's specialized archive infrastructure.*

```
python3 sol_fetchtxs.py
```

1. Analyze Token Accounts (ATAs):

Investigate "Token Account Creation Abuse" and rent exemptions.

```
mysql -h <host> -p < sol_atatable.sql
python3 sol_atafetch.py
python3 sol_failed.py
```

1. Generate Statistics:

```
mysql -h <host> -p < sol_stattable.sql
```

4. Reproducing Paper Results

The figure/table reproduction scripts and their required database configuration are documented in `Measurement/paper_figdata/README.md`. After completing the measurement workflows above and preparing the processed SQL tables, follow that directory-level README to generate the paper figures, tables, and intermediate CSVs.

Notes

- **API Limits:** The measurement scripts are optimized for commercial RPC providers (Alchemy). If using free-tier endpoints, you may need to adjust concurrency settings (e.g., `RPCSIZE`) in the scripts to avoid rate-limiting (HTTP 429).
- **Security:** The `SecurityViolation` scripts are for educational and verification purposes only. Do not run these against production services without explicit authorization.

- **Credentials:** Never commit `config.py` containing your private keys or RPC secrets.
- **Full Version:** For ethical considerations, some mutation/PoC components are sanitized in this release. If researchers need access to the full codebase, please contact us via the paper contact email or request access to the restricted Zenodo artifact.

Artifact Release Channels

The artifact follows the release channels described in the paper's Open-Science section:

1. **Public Zenodo record:** [Zenodo record 20328961](#) contains the sanitized, non-sensitive public artifact, including the x402scope framework, configuration template, execution instructions, measurement code, MariaDB schema/query scripts, and figure/table regeneration code.
2. **GitHub repository:** [HexHive/x402scope](#) mirrors the same non-sensitive public artifact and will be maintained long term.
3. **Restricted-access Zenodo record:** [Zenodo record 20329070](#) contains the full x402scope codebase and sensitive validation records for controlled access.

Sanitization Notice

This public repository is the sanitized, non-sensitive release of the x402scope artifact. It intentionally omits sensitive validation records, exact evaluation target configurations, per-facilitator security-rule pass/fail matrices, HTTP and on-chain evidence logs, adjudication records, and exploit-enabling materials. These materials are provided separately through the restricted-access Zenodo record to support independent assessment while reducing the risk of abuse against live x402 payment infrastructure.

License

The source code in this repository is released under the MIT License.

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